WORKED SOLUTION

STEP III 2022, Question 4

Pure • Infinite Products

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.5, *Viète's Product*: https://beyondstep.keithmcnulty.org/ch02/viete.html

The key idea

Everything rests on the double-angle formula $\sinh 2\theta = 2\sinh\theta\cosh\theta$, applied over and over: each use halves the argument of the sinh and leaves behind a factor of cosh. After n halvings the leftover $\sinh(x/2^n)$ is tiny, and it behaves just like $x/2^n$, so letting $n \to \infty$ turns a finite identity into an infinite product. The last two parts are this product evaluated at well-chosen values of x, together with its trigonometric twin.

Solution

Part (i)

Write $P_n(x) = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)$. We prove that $\sinh x = 2^n P_n(x)\sinh\left(\frac{x}{2^n}\right)$ for every positive integer n.

Base case $n = 1$. By the double-angle formula with $\theta = x/2$, $\sinh x = 2\cosh\left(\frac{x}{2}\right)\sinh\left(\frac{x}{2}\right)$, which is the claim for $n = 1$.

Inductive step. Suppose the result holds for some $n = k$. Using the double-angle formula with $\theta = x/2^{k+1}$,

$$\sinh\left(\frac{x}{2^k}\right) = 2\cosh\left(\frac{x}{2^{k+1}}\right)\sinh\left(\frac{x}{2^{k+1}}\right),$$

so

$$\begin{aligned}\sinh x &= 2^k P_k(x)\sinh\left(\frac{x}{2^k}\right) = 2^k P_k(x)\cdot 2\cosh\left(\frac{x}{2^{k+1}}\right)\sinh\left(\frac{x}{2^{k+1}}\right)\\ &= 2^{k+1}P_{k+1}(x)\sinh\left(\frac{x}{2^{k+1}}\right),\end{aligned}$$

which is the result for $n = k+1$. By induction it holds for all positive integers n.

Deduction. If $x \neq 0$ then $x/2^n \neq 0$, so $\sinh(x/2^n) \neq 0$ (the only zero of sinh is at 0). We may therefore divide the identity by $x\sinh(x/2^n)/2^n$, which gives

$$\frac{\sinh x}{x} \cdot \frac{x/2^n}{\sinh(x/2^n)} = P_n(x),$$

as required.

Part (ii)

For $y \neq 0$, dividing the Maclaurin series by y gives

$$\frac{\sinh y}{y} = 1 + \frac{y^2}{3!} + \frac{y^4}{5!} + \cdots = 1 + y^2 \sum_{r=1}^{\infty} \frac{y^{2r-2}}{(2r+1)!}.$$

? **Ask yourself** Is it enough to say "every term after the first tends to 0"?

Not quite: there are infinitely many terms, so we should bound their sum. For $0 < |y| \leq 1$ we have $y^{2r-2} \leq 1$, so

$$0 \leq \frac{\sinh y}{y} - 1 \leq y^2 \sum_{r=1}^{\infty} \frac{1}{(2r+1)!} = y^2(\sinh 1 - 1),$$

and the right-hand side tends to 0 as $y \to 0$. Hence $\dfrac{\sinh y}{y} \to 1$, and taking reciprocals, $\dfrac{y}{\sinh y} \to 1$.

Now fix $x \neq 0$ and let $n \to \infty$ in the identity from (i). Put $y = x/2^n$; then $y \to 0$, so the factor $\dfrac{x/2^n}{\sinh(x/2^n)}$ tends to 1. The left-hand side therefore tends to $\dfrac{\sinh x}{x}$, while the right-hand side is the partial product $P_n(x)$. So the partial products converge to $\dfrac{\sinh x}{x}$, which is exactly the statement

$$\frac{\sinh x}{x} = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\cdots.$$

Part (iii)

With $x = \ln 2$ we have $\mathrm{e}^{x/2^n} = 2^{1/2^n}$, so

$$\cosh\left(\frac{x}{2^n}\right) = \frac{2^{1/2^n} + 2^{-1/2^n}}{2} = \frac{2^{2/2^n} + 1}{2 \times 2^{1/2^n}} = \frac{1 + 2^{1/2^{n-1}}}{2 \times 2^{1/2^n}},$$

on multiplying top and bottom by $2^{1/2^n}$. For $n = 1$ this gives $\cosh\left(\frac{x}{2}\right) = \dfrac{1+2}{2\sqrt{2}} = \dfrac{3}{2\sqrt{2}}$, and for $n = 2$ it gives

$$\cosh\left(\frac{x}{4}\right) = \frac{1 + 2^{1/2}}{2 \times 2^{1/4}},$$

as required.

On the left of (ii), $\sinh(\ln 2) = \frac{1}{2}\left(2 - \frac{1}{2}\right) = \frac{3}{4}$, so $\dfrac{\sinh x}{x} = \dfrac{3}{4\ln 2}$.

On the right, the partial product of the first N factors is

$$P_N(\ln 2) = \frac{\prod_{n=1}^{N}\left(1 + 2^{1/2^{n-1}}\right)}{2^N \cdot 2^{\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^N}}} = \frac{3}{2} \cdot \prod_{k=1}^{N-1} \frac{1 + 2^{1/2^k}}{2} \cdot \frac{1}{2^{1-1/2^N}},$$

where we have taken out the $n = 1$ factor $(1+2)/2$, re-indexed with $k = n - 1$, and summed the geometric series in the exponent. As $N \to \infty$, $2^{1-1/2^N} \to 2$, and $P_N(\ln 2) \to \dfrac{3}{4\ln 2}$ by (ii). Hence the partial products $\prod_{k=1}^{N-1} \frac{1+2^{1/2^k}}{2}$ converge, and

$$\frac{3}{4\ln 2} = \frac{3}{2} \cdot \frac{1}{2} \cdot \prod_{k=1}^{\infty} \frac{1 + 2^{1/2^k}}{2},$$

and cancelling $\frac{3}{4}$,

$$\frac{1}{\ln 2} = \frac{1+2^{\frac{1}{2}}}{2} \times \frac{1+2^{\frac{1}{4}}}{2} \times \frac{1+2^{\frac{1}{8}}}{2} \times \cdots.$$

Part (iv)

The same argument works word for word with sin and cos, because $\sin 2\theta = 2\sin\theta\cos\theta$ has the same form. Induction gives

$$\sin x = 2^n \cos\left(\frac{x}{2}\right) \cdots \cos\left(\frac{x}{2^n}\right) \sin\left(\frac{x}{2^n}\right).$$

For $0 < x < \pi$ we have $0 < x/2^n < \pi$, so $\sin(x/2^n) \neq 0$ and we can divide as before. The Maclaurin series $\sin y = y - \frac{y^3}{3!} + \cdots$ gives, for $0 < |y| \leq 1$,

$$\left|\frac{\sin y}{y} - 1\right| \leq y^2 \sum_{r=1}^{\infty} \frac{1}{(2r+1)!} = y^2(\sinh 1 - 1) \to 0,$$

so $\sin y/y \to 1$ and, letting $n \to \infty$,

$$\frac{\sin x}{x} = \cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)\cos\left(\frac{x}{8}\right)\cdots.$$

(Alternatively, put $x = \mathrm{i}t$ in (ii), using $\cosh(\mathrm{i}t) = \cos t$ and $\sinh(\mathrm{i}t) = \mathrm{i}\sin t$; but the series argument in (ii) was for real y, so redoing it directly is cleaner.)

Take $x = \frac{\pi}{2}$: the left-hand side is $\dfrac{1}{\pi/2} = \dfrac{2}{\pi}$, and the factors are $\cos\left(\frac{\pi}{4}\right)$, $\cos\left(\frac{\pi}{8}\right)$, $\cos\left(\frac{\pi}{16}\right)$, and so on.

Ask yourself How do we turn $\cos(\pi/8)$, $\cos(\pi/16)$, … into nested square roots?

Use $\cos 2\theta = 2\cos^2\theta - 1$ in the form $2\cos\theta = \sqrt{2+2\cos 2\theta}$, where we take the positive root because $\cos\theta > 0$ for $0 < \theta < \frac{\pi}{2}$. Let $c_n = 2\cos\left(\pi/2^{n+1}\right)$. Then $c_1 = 2\cos\frac{\pi}{4} = \sqrt{2}$, and with $\theta = \pi/2^{n+2}$,

$$c_{n+1} = \sqrt{2+c_n}.$$

So $c_1 = \sqrt{2}$, $c_2 = \sqrt{2+\sqrt{2}}$, $c_3 = \sqrt{2+\sqrt{2+\sqrt{2}}}$, and so on, and the nth factor of the product is $c_n/2$. Therefore

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2+\sqrt{2}}}{2} \times \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \times \cdots,$$

which is Viète's formula of 1593.

Checks

- In (i), put $n = 1$ and $x = 2\theta$: you recover the double-angle formula, as you should.
- In (iii), $\cosh(\frac{1}{2}\ln 2) = \frac{3}{2\sqrt{2}} \approx 1.0607$, and the formula for general n agrees with it at $n = 1$.
- Numerically, the product in (iii) settles quickly: the first 60 factors give $1.442695\ldots$, matching $1/\ln 2 = 1.4426950\ldots$; and the product in (iv) gives $0.6366197\ldots$, matching $2/\pi$. Both were confirmed by computer to about 15 significant figures.

Common slips

- Dividing by $\sinh(x/2^n)$ without noting that it is non-zero because $x \neq 0$.
- Off-by-one errors in (iii): the $n = 1$ factor is $3/(2\sqrt{2})$, not of the form $(1+2^{1/2^k})/2$, and the stray powers $2^{1/2}\cdot 2^{1/4}\cdots$ combine to give a factor of 2, not 1.
- Assuming "$y/\sinh y \to 1$" without any argument from the series, which is what the question asks for.

Beyond STEP

An infinite product $\prod a_n$ is defined as the limit of its partial products, just as an infinite sum is the limit of its partial sums; taking logarithms turns one into the other, so convergence of $\prod(1+u_n)$ is closely tied to convergence of $\sum u_n$. The product for $\sin x/x$ here sits alongside Euler's more famous product $\sin x = x\prod_{k\geq 1}\left(1 - x^2/(k^2\pi^2)\right)$, which leads to $\sum 1/k^2 = \pi^2/6$. For the careful treatment of limits that makes arguments like part (ii) rigorous, read Abbott, *Understanding Analysis*.

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